

TRANSIENT COALITION MEMORY

A sparse, adaptive architecture for certified belief formation

Vision, mechanism, evidence, accomplishments, limitations, and research roadmap

TCM asks a different question: what is the smallest trustworthy coalition of evidence needed to form a belief, and when should that belief be revised?

Prepared for research communication and external review

Version 1.0 research program, including Waves IV–XVI

Executive Summary

Transient Coalition Memory (TCM) is a memory-and-inference architecture designed for environments where facts can change, sources can disagree, feedback arrives late, and computation is limited. Instead of activating every available source and storing every observation permanently, TCM recruits a small coalition, certifies a decision, preserves unused evidence in reserve, and updates trust only when outcomes become known.

The vision

Build an adaptive belief system that remains accurate without paying the cost of exhaustive evidence processing, while retaining enough uncertainty and reserve capacity to recover when the world changes.

What it does

TCM turns many noisy, conflicting reports into a compact, confidence-bearing decision. It learns which sources are useful, which combinations work together, when memory should guide a decision, and when new evidence should override the past.

What it has accomplished

Across the internal benchmark program, TCM reached a high-accuracy, low-compute operating point and outperformed a provenance-graph baseline under matched accounting. Wave XIII formalized the fixed-k comparison and showed that adaptive recruitment sits on the observed accuracy–cost frontier. Real-data stock experiments then exposed a genuine limitation: persistence-heavy data and correlated sources constrain change detection, producing a measurable accuracy–sensitivity frontier rather than a free win.

What it is not

TCM is not a universal truth machine, and current evidence does not establish broad real-world superiority. Its strongest demonstrated claim is architectural: sparse recruitment and reserve evidence can preserve near-maximal accuracy at substantially lower measured inference cost, while making the tradeoffs around change sensitivity explicit.

1. The Problem TCM Is Built to Solve

Most conventional systems make one of two expensive assumptions: either every observation should be retained and reprocessed, or a fixed amount of evidence should always be consulted.

Both assumptions break down when the environment is changing and sources are unevenly reliable.

Environmental stressor	Why it is difficult	TCM response
Changing truth	Yesterday's correct belief can become today's error.	Anchors are retained, but can weaken or be overridden when certified evidence contradicts them.
Conflicting sources	Reports can disagree because of noise, bias, delay, or dependence.	Sources earn reliability through outcomes, and coalitions are judged by their combined evidence.
Delayed feedback	The system must decide before it knows the answer.	Learning is separated from inference; provenance and pending decisions are updated when feedback arrives.
Context dependence	A source can be useful in one regime and poor in another.	Reliability, hazard, and coalition behavior can be conditioned on item or context.
Compute limits	Processing all evidence can be wasteful or impossible.	Adaptive recruitment stops once the decision is sufficiently certified and keeps the remainder in reserve.

2. The TCM Vision

The long-term vision is a belief architecture that behaves less like a database and more like a living evidential system. It should form temporary working coalitions, spend computation only where uncertainty demands it, preserve contradictory evidence rather than erase it, and revise itself without destroying everything it previously learned.

01	Sparse by default	Use the smallest coalition that can support a certified decision.
02	Adaptive rather than fixed	Recruit more evidence only when the current coalition is insufficient.
03	Memory with an escape route	Let prior belief stabilize inference, but never make memory impossible to challenge.
04	Reserve evidence	Keep unused sources available as a low-cost recovery mechanism.

05	Outcome-grounded trust	Update source and coalition reliability from delayed environmental feedback.
06	Honest uncertainty	Expose confidence, calibration, and change-sensitivity tradeoffs rather than hiding them behind a single accuracy number.

3. How TCM Works

TCM separates inference, certification, memory, reserve evidence, and learning. The architecture can be understood as a seven-stage cycle.

1. Receive reports	An item or question receives claims from multiple sources, each with provenance and current reliability estimates.
2. Form an initial coalition	A small subset is recruited rather than activating the full source pool.
3. Aggregate evidence	The coalition combines source support, contradiction, memory anchors, and contextual reliability.
4. Certify or continue	If confidence crosses the decision threshold, recruitment stops. Otherwise, additional reserve sources are activated.
5. Preserve reserve	Evidence that was not needed remains available for contradiction checks, later updates, and efficiency accounting.
6. Commit a transient belief	The decision is stored with confidence, provenance, context, and the coalition that supported it.
7. Learn from delayed feedback	When the environmental outcome arrives, source reliability, coalition behavior, item hazard, and anchor strength are updated.

Core Mechanisms

Adaptive recruitment: The number of activated sources is a decision variable, not a constant. Easy cases stop early; difficult or contradictory cases recruit more evidence.

Certification: A belief is committed only when the coalition has enough weighted support under the model's thresholding rules.

Anchored memory: Past beliefs reduce unnecessary re-computation in stable environments, but create inertia that must be measured and controlled.

Reserve contradiction: Unused sources are not discarded. They provide a pool that can expose disagreement, trigger more recruitment, or support an override.

Reliability learning: Sources and coalitions gain or lose influence according to how well they predict later outcomes.

Per-item hazard memory: Items that often change should receive weaker anchors than items that usually persist.

Conviction override: Strong reliability-weighted consensus can defeat an anchored preference when the current world clearly outweighs memory.

Calibration: Confidence is evaluated separately from accuracy. Post-hoc calibration may be used when raw confidence is systematically under- or over-confident.

4. What TCM Has Accomplished

The research program progressed through a sequence of increasingly demanding benchmark waves. The evidence should be read as a ladder: internal architectural validation first, fixed-policy comparison second, then external stress testing and failure analysis.

Wave	Purpose	Main result	Scientific meaning
XI	Matched benchmark against a provenance graph	Accuracy 0.986 vs 0.968; changed-fact accuracy 0.966 vs 0.952; Brier 0.018 vs 0.025; about half the operations per correct prediction.	The combined cellular-and-gating architecture delivered both higher quality and lower measured compute under the frozen synthetic setting.
XII	Calibration probe	Confirmed that raw TCM confidence was underconfident and that calibration improved probabilistic reporting.	Accuracy and confidence quality must be treated as separate properties.
XIII	Fixed-k Pareto test	Adaptive TCM: 0.9859 accuracy at 9.72 ops/correct. It dominated fixed k=6 and essentially matched k=8 at about 39% lower cost.	Accuracy comes primarily from the cellular core; efficiency comes from adaptive recruitment. Together they occupy the observed Pareto frontier.
XIV–XVI	First real-data stock transfer and forensic analysis	High overall accuracy and low activation cost transferred, but change detection weakened. Source	The external test exposed anchor inertia, effective evidence redundancy, and

Wave	Purpose	Main result	Scientific meaning
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		agreement was highly correlated and the benchmark appeared to contain questionable day-level gold labels.	an accuracy–sensitivity frontier rather than a universal win.
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The Central Engineering Result

TCM can preserve nearly all of the accuracy available from a larger evidence pool while activating only a small fraction of that pool. The adaptive gate is valuable because it controls cost, not because it creates accuracy from nothing.

The Central Scientific Result

The work also identified where TCM fails. In persistent environments with delayed feedback, anchored memory can become a persistence machine. Raising flip sensitivity increases responsiveness but sacrifices overall accuracy at a measurable rate. That tradeoff is now part of the architecture’s claim, not an inconvenient result to conceal.

5. Why TCM Matters

AI systems with expensive retrieval — A model could activate only enough documents, tools, agents, or sensors to certify an answer, reducing inference cost while retaining escalation capacity.

Sensor fusion — Robotics, industrial monitoring, and distributed sensing often contain redundant or correlated reports. TCM provides an explicit mechanism for sparse fusion and delayed reliability learning.

Truth discovery and data integration — Conflicting claims from multiple sources can be resolved without assuming each source is independent or equally trustworthy.

Continual and nonstationary learning — The distinction between stable memory and contradiction-sensitive reserve may help systems retain useful knowledge without becoming unable to revise it.

Human and organizational decision support — TCM offers a model for forming decisions from a temporary coalition of evidence while recording who contributed, what was ignored, and why confidence was sufficient.

6. What Makes TCM Different

Conventional approach	TCM alternative	Practical consequence
Process every source	Recruit until certified	Cost scales with difficulty rather than raw source count.

Store one fused answer	Store belief, confidence, coalition, provenance, and reserve	The system can explain what supported the decision and what contradictory evidence remained.
Use fixed source weights	Learn source and coalition reliability from outcomes	Trust becomes empirical and context-sensitive.
Treat memory as always helpful	Measure anchor benefit and anchor inertia	Stability and responsiveness become tunable, testable properties.
Report one headline metric	Report accuracy, changed-fact accuracy, calibration, activation, and cost	Tradeoffs remain visible and reviewers can see what the system truly accomplishes.

7. Current Limitations and Boundaries

- **Synthetic dependence:** The strongest results were first established in a synthetic environment. The generator may share assumptions with the architecture, so external transfer remains essential.
- **Change detection under persistence:** Anchored memory can suppress rare flips, especially when feedback is delayed and the prior probability of change is low.
- **Correlated sources:** A large report count does not imply a large amount of independent evidence. TCM must estimate or account for source dependence rather than treating all votes as separate.
- **Benchmark quality:** External gold labels can be noisy or contaminated. Claims about model failure or superiority must be accompanied by dataset forensics and sensitivity analyses.
- **Operation counts are model-specific:** Ops/correct is useful for matched comparisons inside the benchmark, but it is not identical to wall-clock speed, energy use, or hardware cost.
- **Calibration is unfinished:** Raw confidence has shown underconfidence in some settings. Calibration must be measured on held-out data and reported transparently.
- **No universal superiority claim:** The present evidence supports a promising architecture and a strong internal Pareto result, not a claim that TCM dominates all memory or truth-discovery systems in every environment.

8. Research Roadmap

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Wave XVII: untouched Weather transfer — Apply the frozen architecture and predefined metrics to a different real source structure without mechanism tuning.

- 2 **Real-data adapter layer** — Create a stable interface for items, contexts, sources, delayed labels, and report provenance so new datasets can be added without rewriting the core.
- 3 **Dependence-aware recruitment** — Estimate source clusters or effective sample size so a correlated block of reports cannot masquerade as independent evidence.
- 4 **Hazard-aware memory** — Continue developing item-level change priors using feedback-only updates, with strict separation between development and held-out evaluation.
- 5 **Wall-clock and resource profiling** — Connect operation accounting to runtime, allocation, and hardware measurements.
- 6 **Ablation and mechanism isolation** — Quantify the separate contributions of cellular inference, adaptive recruitment, reserve evidence, anchoring, hazard memory, and override behavior.
- 7 **External replication** — Release deterministic scripts, environments, raw outputs, and tests so independent researchers can reproduce every headline number.

9. The TCM Claim, Stated Precisely

Transient Coalition Memory is a sparse, adaptive belief architecture that recruits evidence until a decision is certified, preserves unused evidence in reserve, and learns source and memory reliability from delayed outcomes.

Its demonstrated strength is achieving a favorable accuracy–compute tradeoff under matched benchmarks; its central open challenge is maintaining that efficiency while responding reliably to rare changes in correlated, imperfect real-world data.

10. Plain-English Explanation

Imagine fifty people can answer a question, but asking all fifty every time is expensive. TCM starts with a small group. If that group agrees strongly and has a good history, TCM decides. If the answer is uncertain or other evidence pushes back, it brings in more people. It remembers what happened last time, but it also keeps unused voices in reserve so the past cannot become absolute. Later, when the true outcome is known, it updates who was trustworthy and which combinations worked. The goal is simple: reach the right answer with the least unnecessary work, while staying able to change its mind when the world changes.

Conclusion

TCM began as an attempt to build a more efficient memory system. It has become a broader theory of how adaptive systems should form beliefs under uncertainty: temporary coalitions instead of total activation, certification instead of blind accumulation, reserve evidence instead of premature deletion, and outcome-grounded revision instead of fixed trust. The internal results show that this design can be both accurate and computationally sparse. The real-data results show that responsiveness, source dependence, and benchmark integrity are now the decisive scientific frontier. That combination—strong performance, explicit failure analysis, and a testable path forward—is the real accomplishment of the program.

Appendix A. Key Terms

Term	Meaning
Coalition	The subset of sources actively used to form a decision.
Recruitment	The process of adding sources to the active coalition.
Certification	The rule that determines when the current evidence is sufficient to commit a belief.
Reserve	Available but currently inactive evidence that can be recruited later.
Anchor	A prior belief or memory that stabilizes the current decision.
Hazard	The estimated probability that an item’s truth will change.
Override	A mechanism allowing strong current evidence to defeat an anchored prior.
Changed-fact accuracy	Accuracy specifically on cases where the truth differs from its prior state.
Brier score	A measure of probabilistic accuracy; lower is better.
Pareto frontier	A set of solutions where improving one objective, such as accuracy, requires worsening another, such as computation.